



# Drought, heatwave, and fires: The impact on air quality during São Paulo's record-breaking fire season in 2024

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## ABSTRACT

Extreme compound climate events, such as simultaneous heatwaves and droughts, have become more frequent and intense, increasing the risk of forest fires and air quality degradation. Although previous studies have linked compound events to fires in some biomes, a gap remains in understanding their cascading impacts, from climate extremes to fires and, subsequently, to air quality, particularly in densely populated subtropical regions such as São Paulo, Brazil. This study analyzed the evolution of active fires and their relationship with compound drought and heatwave events, as well as their effects on air quality ( $PM_{10}$  and  $O_3$ ), during the exceptional 2024 fire season. The year 2024 experienced one of the state's most severe droughts ( $SPI < -2.0$ ) and the highest temperatures since 1961. This combination favored an unprecedented fire season in which São Paulo recorded 8712 active fires, a 118% increase relative to the historical mean and the highest value since 2003. Nearly 70% of these fires occurred in August (3612) and September (2522), months that registered four compound drought and heatwave events. During these periods,  $PM_{10}$  and  $O_3$  concentrations frequently exceeded WHO guidelines, and a strong positive correlation was observed between the maximum temperature and pollutant levels ( $r = 0.79$  for  $PM_{10}$ ,  $r = 0.87$  for  $O_3$ ). These results underscore the importance of incorporating compound event analysis into environmental monitoring and developing targeted mitigation and adaptation policies. The 2024 season serves as a critical warning of this new climate reality.

## 1. Introduction

Anthropogenic activities are profoundly altering global climate patterns and intensifying weather extremes (Engdaw et al., 2023; IPCC, 2023). The last four decades have been progressively warmer than any other period recorded since 1850, and extreme heat events have become more frequent, longer-lasting, and intense (Freitas et al., 2023; Gao et al., 2023; IPCC, 2023; Zittis et al., 2024; Forster et al., 2025). Between 1970 and 2021, climate change and extreme events contributed to a significant increase in natural disasters, resulting in economic losses totaling billions of dollars and millions of lives lost (WMO – World Meteorological Organization, 2023). Notably, 90% of these fatalities occurred in developing countries, which are poorer and more vulnerable (WMO – World Meteorological Organization, 2023; Santana et al., 2024).

As the frequency of extreme events accelerates, the likelihood of compound extreme events also increases. These events result from the combination of multiple factors and/or hazards that synergistically intensify social and environmental risks. A common example is

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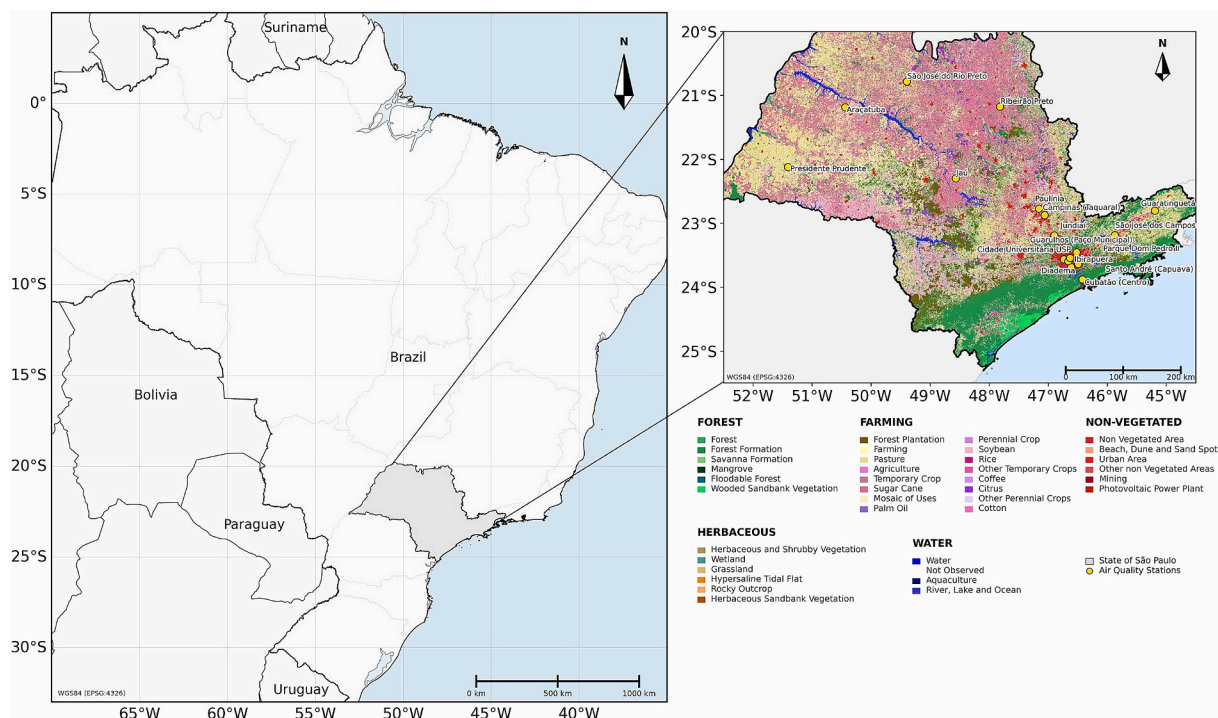
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the co-occurrence of droughts and heatwaves, conditions that promote wildfires through low humidity, high temperatures, and strong winds (Zscheischler et al., 2020; IPCC, 2023). The impacts of a compound extreme event tend to be significantly greater than those caused by isolated extreme events (Zscheischler and Seneviratne, 2017; Ridder et al., 2022).

The relationship between meteorological conditions and fire susceptibility is well-established. Higher temperatures, reduced precipitation, and lower relative humidity increase regional fire susceptibility (Souto et al., 2021). In addition, years marked by more severe droughts and reduced rainfall have been associated with an increased occurrence of forest fires (Rosan et al., 2017; Santos et al., 2019). Wildfires release large quantities of atmospheric pollutants, including particulate matter (PM) and ozone ( $O_3$ ) precursors, severely degrading air quality. From public health to economic stability, the consequences are far-reaching: the increase in pollutant concentrations can aggravate cardiovascular and respiratory diseases; economically, fires reduce agricultural and labor productivity; and ecologically, they disrupt ecosystems and compromise essential natural resources (Sarraf and Mülfarth, 2021; Espindola, 2024; Miguel et al., 2025).

While several studies around the world have focused on the relationship between compound extreme events, such as droughts and heatwaves, and the occurrence of fires (Gouveia et al., 2016; Sutanto et al., 2020; Reddy et al., 2021; Libonati et al., 2022a; Libonati et al., 2022b; Shi et al., 2024), a critical knowledge gap remains. Many of these studies focus on specific biomes, such as boreal or temperate forests, and few have investigated the complete cascading impact, from compound climate event to fire intensification and subsequent air quality degradation, within a densely populated, industrialized subtropical region like São Paulo. Although there are studies that assess air quality and its effects on the health of the population in the state (Santana et al., 2020; Silva et al., 2021a; Wikuats et al., 2023), there is still a lack of research addressing the relationship between compound extreme events, wildfires, and their impacts on air quality. This highlights the need for region-specific analyses to inform targeted adaptation and mitigation plans.

The urgency of this issue in Brazil is stark. Between 1998 and 2024, Brazil recorded approximately 6 million fire outbreaks (CNA, 2024). From 1985 to 2023, it is estimated that approximately 199 million hectares were burned throughout the country (MAPBIOMAS, 2025), highlighting the spatial dimension of the problem. Between 2016 and 2021, economic losses associated with fires in Brazil exceeded USD 208 million (CNM, 2021). In contrast, between June and August 2024 alone, losses in the agricultural sector were estimated at USD 2.89 billion — a value more than 13 times higher than that recorded in the previous period (CNA, 2024). The state of São Paulo stood out as the most impacted during this period, with estimated losses of around USD 552 million in the agricultural sector alone (CNA, 2024). In 2024, the state recorded 8712 fire outbreaks — the highest number since 1998 — with more than 70% concentrated in the months of August and September (INPE – Instituto Nacional de Pesquisas Espaciais, 2025a). Persistent meteorological stability during these months exacerbated pollutant accumulation, causing a sharp decline in air quality (CETESB - Companhia Ambiental do Estado de São Paulo, 2025). The convergence of these factors, such as extreme fire activity, dense urban populations, and industrial emissions, amplifies the public health risks and underscores the urgent need for a targeted investigation.



**Fig. 1.** Study area: a) location of São Paulo state in Brazil and b) detailed map of the state of São Paulo with land use and land cover data from MapBiomas. The yellow dots represent the location of CETESB air quality stations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The state of São Paulo is home to the largest population in the country, with 44,411,238 inhabitants, of which more than 96% reside in urban areas (IBGE, 2022). In addition, São Paulo is also responsible for the largest Gross Domestic Product (GDP) in the country, totaling USD 536,143.26 million (IBGE, 2021; Cachola et al., 2022). This convergence of high population density, industrial activity, and intense vehicular emissions creates a unique hotspot of vulnerability, where the health and environmental risks of wildfire-induced pollution are amplified. Given this critical context, this study aims to: (i) examine the annual and monthly history of fire outbreaks in the state of São Paulo; (ii) identify the occurrence of compound drought and heatwave events from August to September 2024; (iii) correlate these compound events with the number of fire outbreaks; and (iv) assess the cascading influence on air quality ( $\text{PM}_{10}$  and  $\text{O}_3$ ).

## 2. Material and methods

### 2.1. Study area

The state of São Paulo, located in the Southeast region of Brazil (Fig. 1), has an area of approximately 249,000 km<sup>2</sup>, representing 2.9% of the national territory (CETESB - Companhia Ambiental do Estado de São Paulo, 2024; IBGE, 2024). Composed of 645 municipalities, the state of São Paulo stands out for being home to the largest population in the country, the largest vehicle fleet — with 34,332,818 units, representing 27.7% of the national total — and the largest number of companies in the industrial sector, totaling 146,312, equivalent to 24.7% of the Brazilian total (CNI, 2023; IBGE, 2024). This scenario, combined with high economic development, has contributed significantly to the deterioration of air quality (CETESB - Companhia Ambiental do Estado de São Paulo, 2024).

An additional factor that contributes to poor air quality in the state of São Paulo is the occurrence of forest fires. To mitigate the environmental impacts of wildfires, the state implemented the forest fire prevention and fighting system known as “Operação São Paulo Sem Fogo” (Operation São Paulo Without Fire). The main objective of this program is to reduce fire outbreaks and minimize greenhouse gas emissions resulting from wildfires (CETESB - Companhia Ambiental do Estado de São Paulo, 2024). The climate, characterized by rainy summers and dry winters associated with the South American Monsoon System (SAMS; Reboita et al., 2012) makes the region highly susceptible to wildfires during the late dry season.

### 2.2. Data

Active fires from Moderate-resolution Imaging Spectro-radiometer (MODIS) sensor aboard Earth Observation System - AQUA satellite from 2003 to 2024 collection 6 fire product (Giglio et al., 2016) processed by the National Aeronautics and Space Administration (NASA) were used. Daily active fires are detected by the 4.0  $\mu\text{m}$  MODIS channel with a spatial resolution of 1.0 km. The daily afternoon overpass of AQUA satellite, known as the reference satellite (AQUA\_M-T), is reprocessed and provided by Wildfire Monitoring Program from the National Institute for Space Research (INPE – Instituto Nacional de Pesquisas Espaciais, 2025b; Setzer et al., 2019; Setzer et al., 2021). Each active fire corresponds to a pixel that contains one or more actively burning fires at the time of satellite overpass. The detection algorithms compare the temperature of a potential fire with the temperature of the surrounding land cover; if the temperature difference exceeds a predefined threshold, the potential fire is confirmed as an active fire, or “hot spot” (Giglio et al., 2016). This database is ideal for compiling time series over the years and thus enables the analysis of trends in the number of outbreaks for the same regions and between regions in periods of interest (INPE – Instituto Nacional de Pesquisas Espaciais, 2025c).

Aerosol optical depth (AOD) over land retrieved in the MODIS green band (0.55  $\mu\text{m}$ ) was also used. MCD19A2 V6.1 data product is a MODIS TERRA and AQUA combined Multi-angle Implementation of Atmospheric Correction (MAIAC) Land AOD gridded Level 2 product produced daily at 1 km spatial resolution (Lyapustin and Wang, 2022). AOD dataset is provided by NASA LP DAAC at the USGS EROS Center, and are available and processed by Google Earth Engine (Google Earth Engine, 2025; Gorelick et al., 2017).

Daily precipitation and maximum temperature (Tmax) data for the period spanning 1980 to 2024 were obtained from the Climate Prediction Center (CPC), produced by the National Oceanic and Atmospheric Administration (NOAA, n.d). This gridded dataset (0.5° resolution) is generated through the interpolation of over 30,000 global meteorological stations (Chen et al., 2008; Sun et al., 2018). For this study, we extracted and calculated the mean of the data for all grid points located within the administrative boundaries of São Paulo state to create a single representative time series for regional analysis.

Hourly data for  $\text{O}_3$  and  $\text{PM}_{10}$  concentrations in the São Paulo state from 2003 to 2024 were obtained from the QUALAR website (Air Quality) of the Environmental Company of the State of São Paulo (CETESB - Companhia Ambiental do Estado de São Paulo, 2025). This study included measurements of 17 air quality monitoring stations distributed throughout the state, as detailed in the Supplementary Material and shown in Fig. 1. As shown in Table S1, only the *Cidade Universitária USP* station did not present  $\text{PM}_{10}$  data. This absence did not impact the results, since the analysis considered the spatial mean in the state, thus mitigating any individual monitoring gaps. The selection of these pollutants is justified by the fact that both frequently exceed state air quality standards in São Paulo (Gavidia-Calderón et al., 2024), in addition to being directly associated with emissions from fires (Coelho et al., 2023), contributing significantly to the degradation of air quality.

### 2.3. Assessment of fire outbreaks

To evaluate the temporal-spatial behavior of wildfires, active fires that occurred in Brazil, and specifically in São Paulo state, between 2003 and 2024 were selected. Subsequently, the occurrence of active fires was interpolated to a horizontal grid of 20-km spatial resolution using the bi-linear interpolation method. In this method, each pixel's fire count corresponds to the total number

of detections falling within its 20-km cell, providing a standardized basis for spatial comparison. This spatial smoothing helps highlight regional patterns while reducing noise associated with highly localized detections. To compare the number in São Paulo state with those in Brazil, we evaluated the total annual temporal series of active fires for both domains. In addition, an analysis of annual and monthly spatial anomalies of active fires for São Paulo state was conducted to identify which regions of São Paulo experienced the above the mean wildfire activity relative to the long-term climatology. This approach highlights both persistent hotspots and years with atypical fire behavior. Consistently, the spatial annual anomaly of AOD 550 nm was also evaluated. Evaluating AOD anomalies alongside fire anomalies allows us to investigate whether aerosol load increases are spatially coherent with wildfire occurrence. Together, these diagnostics provide insight into the linkage between fire activity and atmospheric aerosol variability in the region.

To evaluate long-term wildfire trends in São Paulo state, Sen's slope estimator was applied to the annual active fire series for the period 2003–2024 (Sen, 1968). Statistical significance was assessed using the non-parametric Mann–Kendall test at the 95% confidence level ( $p < 0.05$ ; Mann, 1945; Kendall, 1975). Positive Sen's slope values indicate increasing wildfire trends, while negative values represent decreasing trends over time.

#### 2.4. Analysis of long-term drought conditions and compound events

The long-term drought conditions in the São Paulo State were assessed through the 12-month Standardized Precipitation Index (SPI-12), as proposed by McKee et al. (1993). This index is based exclusively on historical precipitation series to determine dry and rainy periods (Fernandes et al., 2009). This index quantifies precipitation anomalies over a 12-month period, which is effective for monitoring meteorological droughts and their impacts on reservoir levels and river flows (McKee et al., 1993; WMO – World Meteorological Organization, 2012). We calculated SPI-12 values for the study period to establish whether the peak fire season occurred under pre-existing long-term dry conditions. The SPI index is classified as: extremely wet ( $\geq 2.0$ ), very wet (1.5 to 1.99), moderately wet (1.0 to 1.49), near normal ( $-0.99$  to  $0.99$ ), moderately dry ( $-1.0$  to  $-1.49$ ), severely dry ( $-1.5$  to  $-1.99$ ) and extremely dry ( $\leq -2.0$ ) (McKee et al., 1993).

For the daily analysis from August to September 2024, compound events were identified based on the simultaneous occurrence of a short-term dry spell and a heatwave. Dry spells were defined as periods with at least five consecutive days of precipitation below 1 mm/day, following the methodology proposed by Olmo et al. (2022). A heatwave was identified based on the calculation of Tmax 90th percentile (p90; Wilks, 2019). The p90 threshold was calculated for each month (August and September) using a long-term time series (1980–2024) to ensure a robust and standard comparison (Russo et al., 2014). Hence, a specific threshold was obtained for each month. Although there are different methods to identify heatwaves, in this study, these events were classified as periods in which Tmax values exceeded p90 thresholds for at least three consecutive days (Russo et al., 2014; Diniz et al., 2020). An event was classified as a compound extreme only on the days when both a dry spell and a heatwave event occurred simultaneously.

#### 2.5. Air quality

Hourly concentrations of O<sub>3</sub> and PM<sub>10</sub> were obtained from 17 CETESB air quality monitoring stations across the State of São Paulo. For O<sub>3</sub>, an 8-h moving average was calculated for each day, resulting in 17 daily values obtained from consecutive hourly series. Then, the highest of these values was selected to represent the maximum daily moving average. This procedure follows standard air-quality guidelines and ensures that short-term peaks in O<sub>3</sub> concentration are appropriately captured. For PM<sub>10</sub>, daily mean concentrations were computed from the 24-hourly records. Data quality control followed WHO recommendations, and means with at least 75% valid records were calculated and included in the analysis. These validated daily data were then used to derive monthly means for the 2024 study period and to establish a long-term climatological mean for anomaly analysis.

Initially, air quality conditions were evaluated by comparing the daily maximum 8-h O<sub>3</sub> concentrations and daily PM<sub>10</sub> means against the World Health Organization Air Quality Guidelines (WHO – World Health Organization, 2021). Additionally, monthly means were compared with long-term climatology to identify air quality high concentrations episodes. Results were presented through time series and anomaly plots for August and September 2024, corresponding to the peak of fire activity.

A Pearson correlation analysis was performed to quantify the relationship between atmospheric pollutants (O<sub>3</sub> and PM<sub>10</sub>), maximum temperature (Tmax), fire outbreaks, and precipitation (Wilks, 2019). The statistical significance of the correlations was evaluated at a  $p$ -value  $< 0.05$ . In addition to the statewide analysis, correlation matrices were calculated for four representative cities (Jaú, Jundiaí, Ribeirão Preto and São José do Rio Preto), selected based on their relevance in the hotspot analysis and availability of air quality observations.

### 3. Results and discussion

#### 3.1. Historical analysis of active fires and their drivers

In 2024, Brazil experienced a 19% increase in active fires compared to the historical mean, making it the most active fire year since 2011 (Supplementary Fig. S1). Within this national context, the state of São Paulo emerged as a critical hot spot, with a staggering 118% increase, totaling 8712 active fires (the highest value since 1998), and approximately 70% of these records occurred in August and September (INPE – Instituto Nacional de Pesquisas Espaciais, 2025a). These values contrast sharply with those of 2022 and 2023, when the number of active fires in São Paulo decreased by 59 and 58%, respectively (Supplementary Fig. S1).

To complement the historical analysis, a spatial trend assessment based on Sen's slope estimator and the Mann–Kendall significance

test was performed (Fig. 2). The results revealed predominantly negative long-term wildfire trends across large portions of São Paulo state, particularly in the southern and southeastern regions, where statistically significant decreases ( $p < 0.05$ ) were identified. However, localized positive trends were observed in parts of the northern sector, highlighting the spatial heterogeneity of wildfire dynamics. Despite these long-term decreasing trends, the extreme drought and record-breaking temperatures observed in 2024 created exceptionally favorable conditions for wildfire ignition and spread, contributing to the unprecedented fire activity recorded during the study period.

The unprecedented fire activity in 2024 probably was not driven by a single factor, but rather a convergence of severe climate pre-conditioning and persistent human pressure. From a climate perspective, 2024 proved to be an atypical year, with São Paulo facing its worst drought (200 to 400 mm below the mean) in seven decades and the highest temperatures (2 °C above the mean) since 1961, a scenario that was exacerbated by several heatwaves throughout the year (Marengo et al., 2025).

Drought conditions played a key role in this pattern, as observed in the Pantanal in 2019 and 2020 (Marengo et al., 2021; Garcia et al., 2021). To quantify the long-term precipitation deficit in the state of São Paulo, the SPI-12 (Fig. 3) was calculated, an index used to identify hydrological drought conditions throughout the historical series from 1980 to 2024. The SPI-12 confirmed that 2024 was one of the driest years in the historical series, with SPI values below  $-2$  (indicating extreme drought conditions) throughout much of the state (Fig. 3). The hydrological crisis also affected critical water sources, such as the Cantareira System, responsible for supplying the metropolitan region of São Paulo, as reported by CEMADEN (CEMADEN – Centro Nacional de Monitoramento e Alertas de Desastres Naturais, 2024). This prolonged and severe drought created ideal conditions for wildfire ignition and propagation (Santos et al., 2019), as this phenomenon reduces soil and vegetation moisture, increases the amount of available fuel (Cui et al., 2023) and, consequently, results in larger, more severe fires with greater plant mortality (Moss et al., 2024).

The year 2021 presented alarming SPI values, with indices below  $-3$  (Fig. 3). Despite registering a more intense drought, 2021 did not surpass 2024 in terms of impacts, since the latter combined water deficit with extreme heat, being the hottest year since 1961 (Marengo et al., 2025). These conditions favored the ignition and spread of wildfires on a record scale. Of the 2600 active fires registered in São Paulo state between August 22 and 24, 81% occurred in agricultural areas (Garrido, 2024). On the 23rd, the number of active fires increased from 25 to 1886 in less than two hours, indicating a possible intentional action and the great magnitude of these events (Kelley et al., 2025).

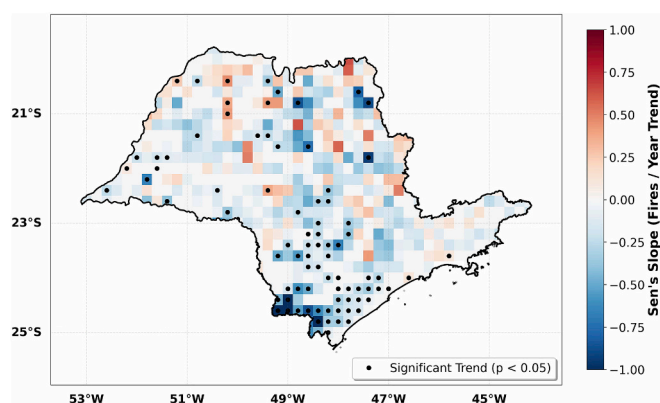
Compounding these natural factors was the significant influence of human activity. Between January and mid-September 2024, the *Operação São Paulo Sem Fogo* recorded 2392 fire occurrences. Enforcement actions resulted in 420 environmental violation reports covering over 107,000 ha, with fines totaling more than USD 4.7 million (SÃO PAULO (Estado), 2024a). Intentional fires were a notable contributor, with at least 12 arrests made for this crime in São Paulo alone (Vieira et al., 2024).

These numbers demonstrate that human actions play a central role in the occurrence of fires, whether intentionally or accidentally. The controlled use of fire, within safety standards and conditions, can function as a soil management practice, bringing benefits such as pest elimination and nutrient supply through ash (Silva et al., 2021b; de Souza and de Lima, 2025). This technique, permitted by law, requires prior knowledge of local conditions and technical supervision to prevent the fire from getting out of control (de Sousa et al., 2024). However, when fire is used uncontrollably, whether by human action or natural causes, it can result in devastating consequences, with significant environmental, social, and economic damage (Silva et al., 2021b; Vieira et al., 2024).

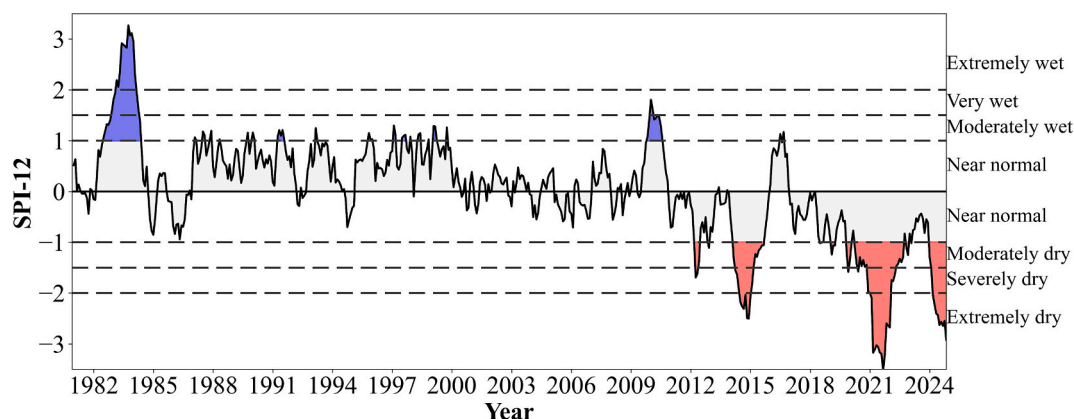
However, although Brazil has strict laws, enforcement faces persistent challenges, including impunity due to difficulties in identifying offenders, insufficient surveillance in remote areas, and slow fine collection (Vieira et al., 2024). These gaps allow illegal activities to continue, magnifying the damage to both the environment and public health.

### 3.2. Spatiotemporal dynamics of the 2024 active fires

The annual anomaly of active fires (Fig. 4) reinforces the severity of the 2024 season, pinpointing the northern and northeastern



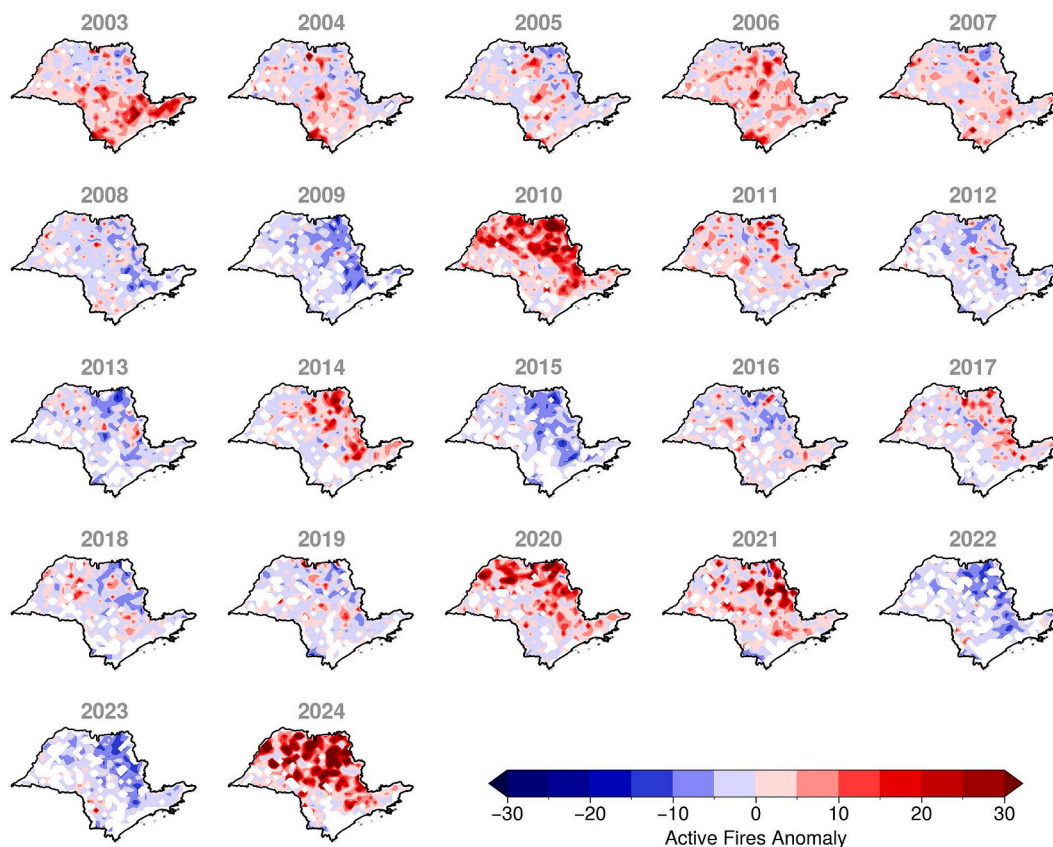
**Fig. 2.** Magnitude of the spatial trend (Sen's slope; fires/year) of active fires in São Paulo between 2003 and 2024. Black dots indicate statistical significance ( $p < 0.05$ ).



**Fig. 3.** SPI-12 monthly averages for São Paulo state from 1982 to 2024. Red (blue) colour represents the dry periods (humid periods). The dashed lines represent the SPI levels from extremely dry (below  $-2$ ) to extremely wet (above  $+2$ ). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

mesoregions as the most impacted areas, represented by the darker shades of red, indicating anomalies exceeding 30 hotspots. These areas are traditionally drier, with high vegetation stress and intensive agricultural activity focused on sugarcane cultivation, a crop in which the state holds a national leadership position (Ogura et al., 2022; Zheng et al., 2022). The mechanized harvesting of this crop leaves a layer of dry straw on the soil surface, which acts as a fine fuel, increasing the risk of fire ignition and spread (Cardoso et al., 2013), and thus making these areas more vulnerable during dry periods.

The vast majority of the year's fire activity was concentrated in August and September. As shown in Fig. 5, these two months recorded the largest positive anomalies (up to 10 active fires), with the northern and northeastern portions of the state, the most affected areas. The Monthly Monitoring and Risk Bulletins for Burning and Forest Fires, published by the National Institute for Space



**Fig. 4.** Annual (from 2003 to 2024) anomalies of active fires for São Paulo state with spatial resolution of 20 km.

Research (INPE), reported that during August and September 2024, fire outbreaks was exceptionally higher than in 2023, increasing from 352 to 3612 (a 926% increase) in August and from 375 to 2522 (a 573% increase) in September. These bulletins also indicated the northern and northeastern portions of the state as the areas at greatest risk for fires (INPE – Instituto Nacional de Pesquisas Espaciais, 2024a; INPE – Instituto Nacional de Pesquisas Espaciais, 2024b), corresponding to the regions with the highest accumulations observed in Fig. 5.

An analysis of the complete historical monthly record from 2003 to 2024 further confirms the exceptional nature of this period. Fig. 6 highlights that August and September 2024 dramatically exceeded the values observed in any other year from 2003 to 2024 (3612 and 2522 events/month, respectively). These months were also well above the climatology (2558 and 1538 events/month, respectively), represented by the dashed line, confirming them as exceptional periods for fire outbreaks. The *Operação São Paulo Sem Fogo* program is divided into three annual phases: green, yellow, and red, with the latter being the most critical. Under normal conditions, August and September are usually classified in the red phase due to a higher fire risk. However, in 2024, the severity of the situation led the state government to create a crisis cabinet and declare a 180-day state of emergency in 45 municipalities. This extreme measure was necessary in response to the record number of fires recorded in August, as evidenced by the highest peak in Fig. 6 (SÃO PAULO (Estado), 2024b).

This scenario highlights that, even with satellite monitoring carried out by INPE and actions implemented by state governments, controlling wildfires in Brazil remains a major challenge due to resource limitations, weak monitoring practices, and the cultural use of fire as an agricultural practice (Botelho et al., 2025). Climate change intensifies conditions conducive to the occurrence and spread of fires (Veras, 2024). Given this reality, it is essential to review and improve existing public policies, such as the *Operação São Paulo Sem Fogo* program, incorporating approaches that consider the new climate reality (Veras, 2024) and the use of predictive models to reduce risks and anticipate critical periods (Melo et al., 2024).

### 3.3. Compound events, active fires peaks, and air quality degradation

Fig. 7 illustrates the spatial relationship between drought, extreme heat, and wildfire activity during the peak of the 2024 fire season, showing the distribution of dry days (Prec < 1 mm), hot days (Tmax > P90), and active fires during August and September 2024. The results revealed a strong spatial correspondence between regions affected by persistent precipitation deficits, elevated

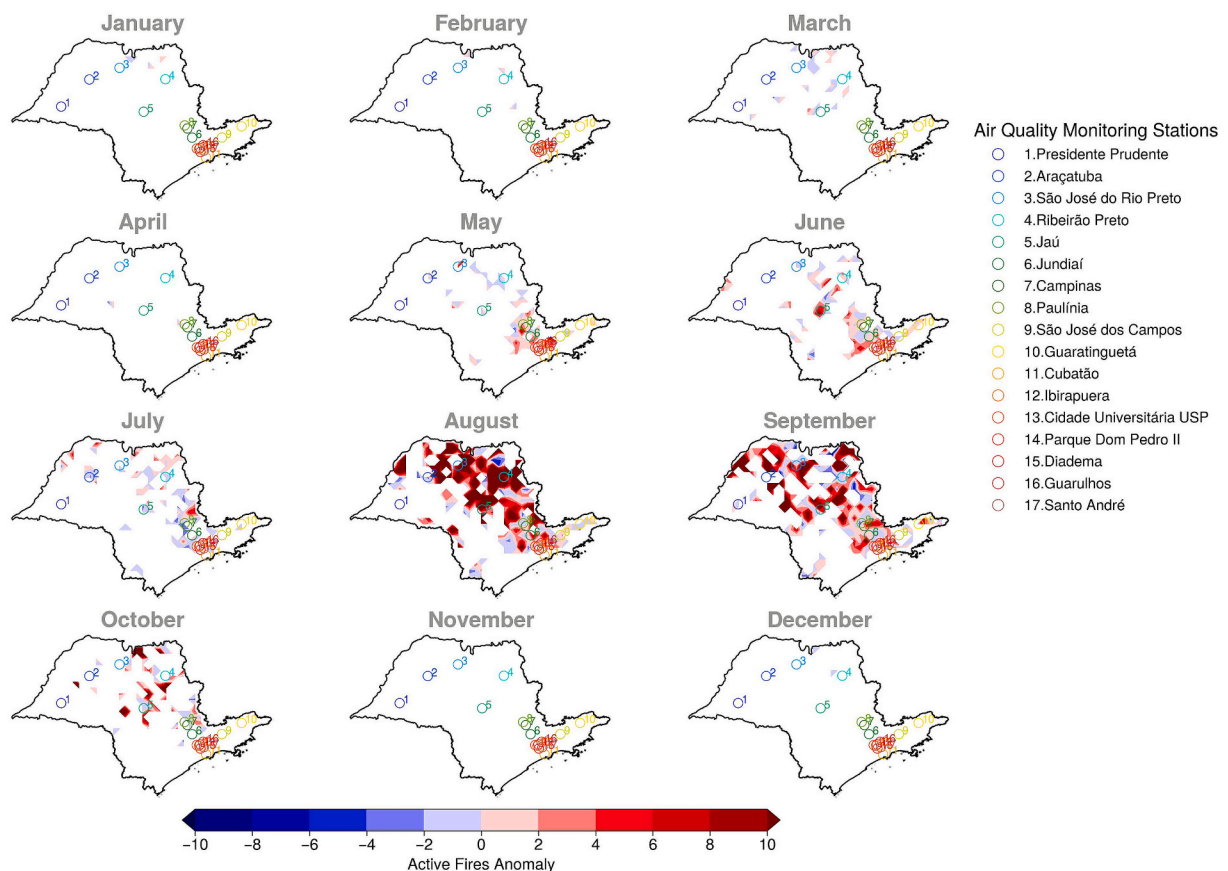
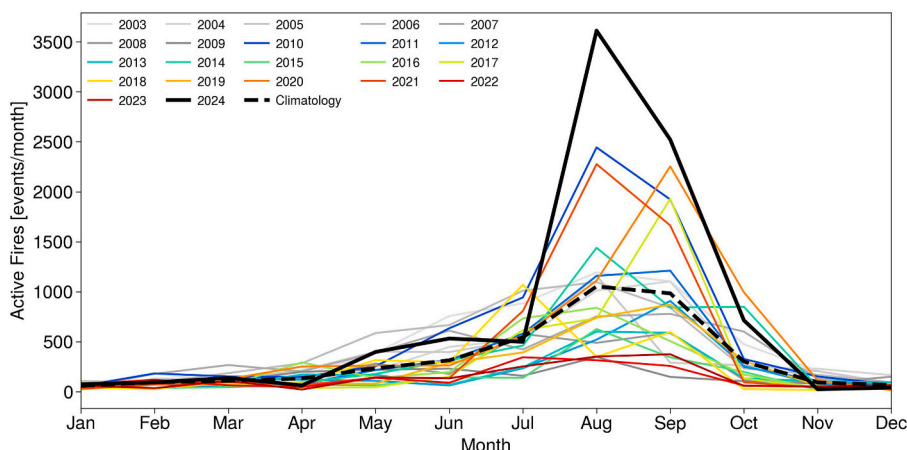


Fig. 5. Monthly anomalies of active fires São Paulo state in 2024. The circles indicate the locations of CETESB stations used in this study.

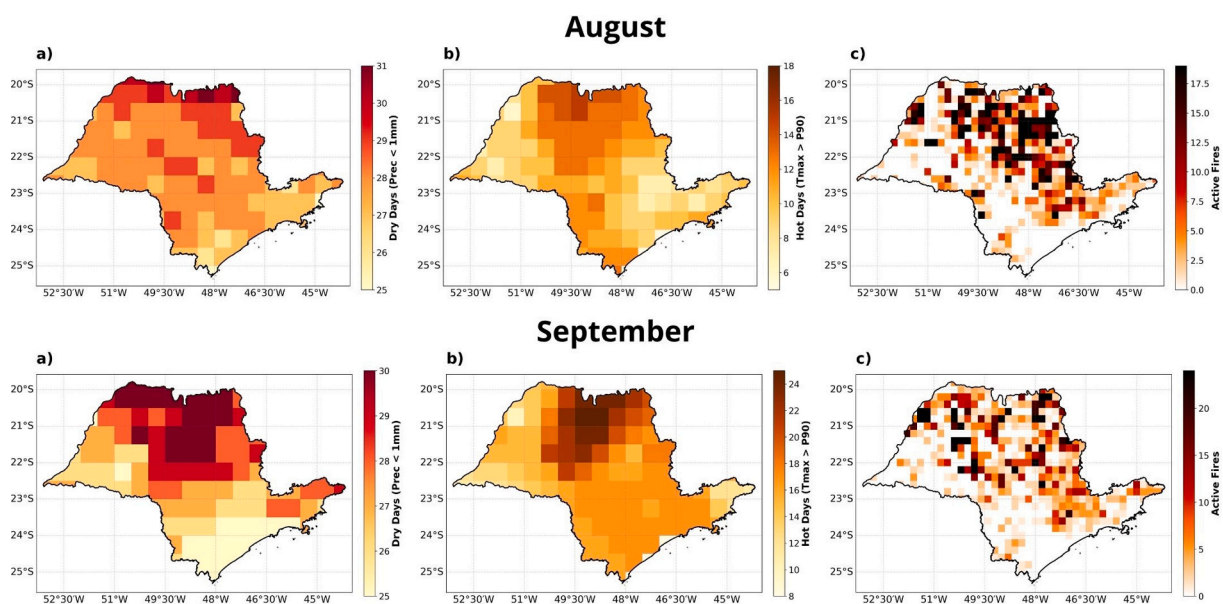


**Fig. 6.** Monthly accumulation of active fires (colored lines) and the corresponding climatology (dashed line) for São Paulo state in the period from 2003 to 2024. The active fires for 2024 are highlighted in black solid line.

frequencies of extreme temperature days, and high wildfire occurrence. In both months, the northern, northwestern, and central sectors of São Paulo exhibited the highest concentration of dry days and hot days, coinciding with the regions most affected by wildfire activity.

These regions also correspond to areas where sugarcane cultivation has expanded substantially over recent decades, particularly in western São Paulo, predominantly through the replacement of pasture areas (Lourenzani and Caldas, 2014; Camara and Caldarelli, 2016). Under the extreme drought and heat conditions observed in 2024, these highly modified agricultural landscapes likely became more susceptible to fire ignition and propagation, contributing to the amplification of wildfire activity across São Paulo state. Ogura et al. (2022) identified the northwestern and western sectors of São Paulo as the regions most susceptible to environmental impacts associated with sugarcane production, including illegal burning and air quality degradation. Under the extreme drought and heat conditions observed in 2024, these highly modified agricultural landscapes likely became even more susceptible to fire ignition and propagation, reinforcing the strong temporal relationship observed between compound climate extremes, wildfire peaks, and atmospheric pollution during August and September 2024.

The temporal evolution of meteorological variables, active fires, and air pollutant concentrations ( $PM_{10}$  and  $O_3$ ) during August and September 2024 reveals a clear and immediate relationship between compound climate extremes and environmental degradation (Fig. 8). The shaded areas correspond to compound extreme events, defined as periods when heatwaves and droughts occurred



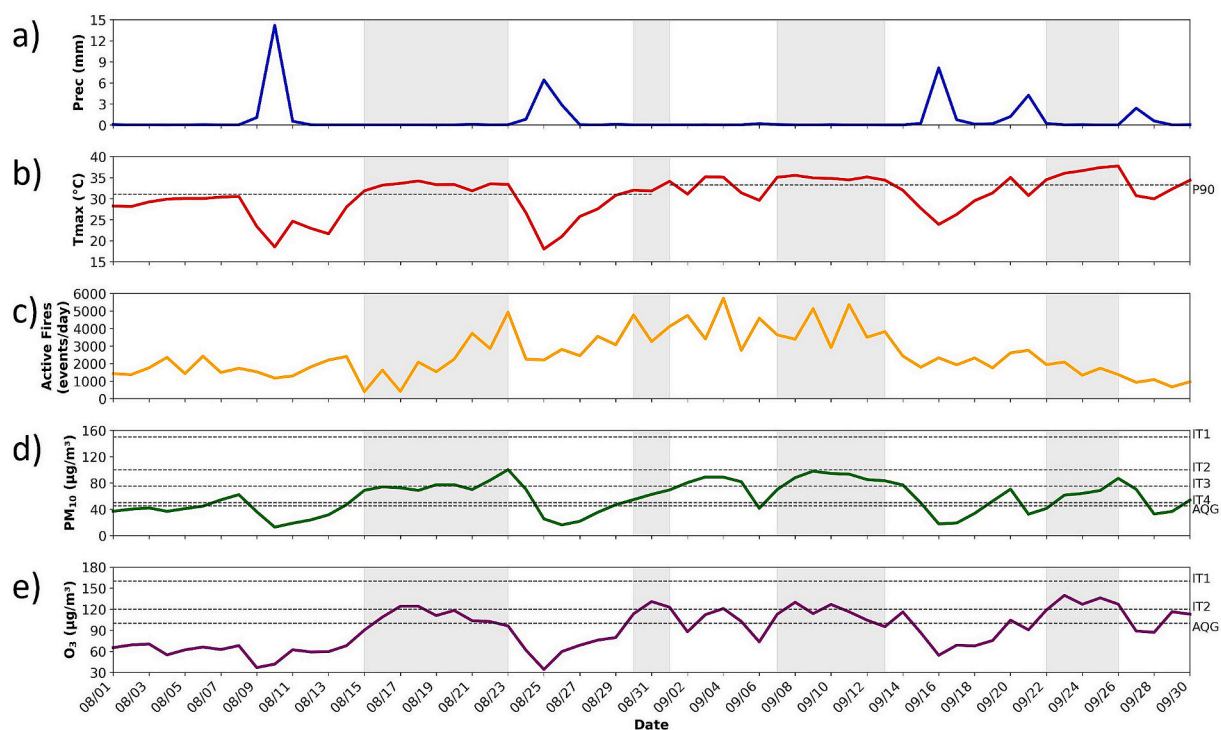
**Fig. 7.** Monthly spatial analysis for August and September 2024 for a) dry days (precipitation < 1 mm), b) hot days (maximum temperature > 90th percentile), and c) number of active fires.

simultaneously. These events occurred between August 15 and 23, August 30 and September 1, September 7 and 13, and September 22 and 26. The compound events consistently coincided with the most pronounced peaks in active fires (Fig. 8c). This strong temporal alignment indicates that high temperatures, combined with a prolonged lack of precipitation, resulted in decreased surface water resources, reduced soil moisture, increased vegetation stress, and decreased fuel moisture (Jiménez et al., 2024; Bourgoïn et al., 2025). Fuel moisture content, which represents the amount of water in plant biomass, is a critical variable for estimating the risk of fire ignition and spread (dos Santos et al., 2022). Under these conditions, any ignition source can generate rapid and intense spread, confirming the relationship between compound heat and drought events and increased fire risk (Sutanto et al., 2020).

This surge in fire activity directly translated into a severe deterioration of air quality. During these compound events, PM<sub>10</sub> concentrations (Fig. 8d) surpassed the WHO Interim Target 2 (IT-2 - 100 µg/m<sup>3</sup> for PM<sub>10</sub>), exceeding the guideline value recommended by the Air Quality Guidelines (AQG), thus posing health risks (WHO – World Health Organization, 2021). Similarly, O<sub>3</sub> concentrations (Fig. 8e) also exceeded IT-2 values (120 µg/m<sup>3</sup> for O<sub>3</sub>) during several compound event periods, highlighting the link between high temperatures, fire emissions, and photochemical formation of O<sub>3</sub> (Lu et al., 2019; Pan and Faloona, 2022). Sarra and Müllfarth (2021) found that, during heat waves, O<sub>3</sub> and particulate matter levels tend to rise, corroborating the results presented here. Higher temperatures accelerate atmospheric photochemical reactions, increasing O<sub>3</sub> concentrations (Du et al., 2024), and forest fires favor the increase in PM<sub>10</sub> in the atmosphere. Under stable conditions, these pollutants are more difficult to disperse, resulting in deteriorating air quality (Gao et al., 2020; Hu et al., 2022).

To statistically validate this observed relationship, a Pearson correlation analysis was performed. The results showed a strong positive association between Tmax and atmospheric pollutants, with coefficients of  $r = 0.79$  for PM<sub>10</sub> and  $r = 0.87$  for O<sub>3</sub>. A high correlation between PM<sub>10</sub> and O<sub>3</sub> ( $r = 0.71$ ) was found, as high temperatures and dry conditions are conducive to the simultaneous increase in both. These conditions favor the occurrence of fires, which release large amounts of PM<sub>10</sub> and trace gases, such as Volatile Organic Compounds (VOCs) and nitrogen oxides (NOx), formed during the biomass combustion process (de Moura et al., 2024; Palmeiro-Silva et al., 2025). In the presence of solar radiation, these gases trigger photochemical reactions that increase O<sub>3</sub> concentrations (de Moura et al., 2024). Thus, fires act as a common source of these pollutants, justifying the observed positive correlation. Notably, the highest correlation of wildfires was observed with PM<sub>10</sub> ( $r = 0.43$ ), indicating that the occurrence of fires is an important source of particulate matter pollution.

PM<sub>10</sub> concentrations were more directly associated with wildfire emissions, since biomass burning releases large quantities of primary particulate matter into the atmosphere. In contrast, O<sub>3</sub> formation involves nonlinear photochemical processes driven by precursor gases such as NOx and VOCs, in combination with solar radiation, high temperatures, and atmospheric stability. Therefore, although wildfire emissions may contribute to O<sub>3</sub> precursor availability, O<sub>3</sub> concentrations are also strongly controlled by meteorological conditions favorable to photochemical production (Lu et al., 2019; Pan and Faloona, 2022). In highly urbanized regions such as



**Fig. 8.** Daily analysis for August and September 2024 for a) precipitation (mm), b) maximum temperature (°C), c) active fires (events/day), d) PM<sub>10</sub> (µg/m<sup>3</sup>) and e) O<sub>3</sub> (µg/m<sup>3</sup>). The shaded areas represent the compound events. The dashed lines indicate the p90 and the reference values for interim air quality averages defined by the WMO.

São Paulo, vehicular and industrial emissions also play an important role in  $O_3$  precursor availability, contributing to the complexity of  $O_3$  formation dynamics during extreme events. This result is consistent with several global studies that have identified biomass burning as a major driver of extreme  $PM_{10}$  episodes (Tarín-Carrasco et al., 2021; Cantamissa et al., 2024; De Souza et al., 2024; Tajudin et al., 2024). On the other hand, precipitation showed a negative correlation with all variables, confirming its role as a mitigating factor for air pollution.

To assess the spatial consistency of these relationships, additional correlation matrices were calculated for four representative cities in São Paulo state: Ribeirão Preto, São José do Rio Preto, Jaú, and Jundiaí, which were also considered in the hotspot analysis presented in Section 3.4 (Supplementary Figs. S3–S6). Overall, the results consistently showed strong positive correlations between  $T_{max}$  and both  $PM_{10}$  and  $O_3$  concentrations, particularly in Jaú, Jundiaí, and São José do Rio Preto. Relatively strong correlations between  $PM_{10}$  and  $O_3$  were also observed in Jundiaí and Ribeirão Preto, suggesting the combined influence of atmospheric stagnation and photochemical processes on air quality degradation. In contrast, active fires generally showed weaker correlations with  $O_3$ , reinforcing that  $O_3$  concentrations were more strongly associated with meteorological conditions favorable to photochemical formation than directly with wildfire occurrence.

This direct link between compound events, fire peaks, and ground-level pollution is further corroborated by a long-term analysis of regional atmospheric aerosol loading using Aerosol Optical Depth (AOD) data at 550 nm (Fig. 9). The periods with positive anomalies (represented in shades of red) indicated high AOD values, associated with higher concentrations of aerosols in the atmosphere, from vehicle, industrial, and fire emissions (Kumar et al., 2007; Kumar, 2010; Yang et al., 2019). The annual AOD anomaly unequivocally identifies 2024 as an extreme outlier, marked by an intense positive anomaly covering a large part of the state, which indicates an exceptionally high concentration of atmospheric aerosols.

A comparison with Figs. 3 and 4 reveals a coherent pattern: the years 2020 and 2021, which had fire activity above the mean and were classified as extremely dry by the SPI-12, also show positive AOD anomalies. In contrast, 2022 and 2023, with fewer fires, show near-zero or negative AOD anomalies. Therefore, a strong relationship is observed between drought conditions, fire occurrence, and the optical parameter. In drier years with a greater number of fires, AOD anomalies were high, while in wetter years with fewer fires, these anomalies remained low or negative. Thus, AOD can be interpreted as the final atmospheric result, representing the total load of emitted and transported aerosols. These results corroborate those of Ramos and de Miranda (2023), who also observed higher AOD values during the dry season. The fact that 2024 simultaneously exhibited the highest fire count since 2003, extreme drought conditions, and the most intense AOD anomaly of the entire 2003–2024 period provides strong evidence of the severity of the event, in

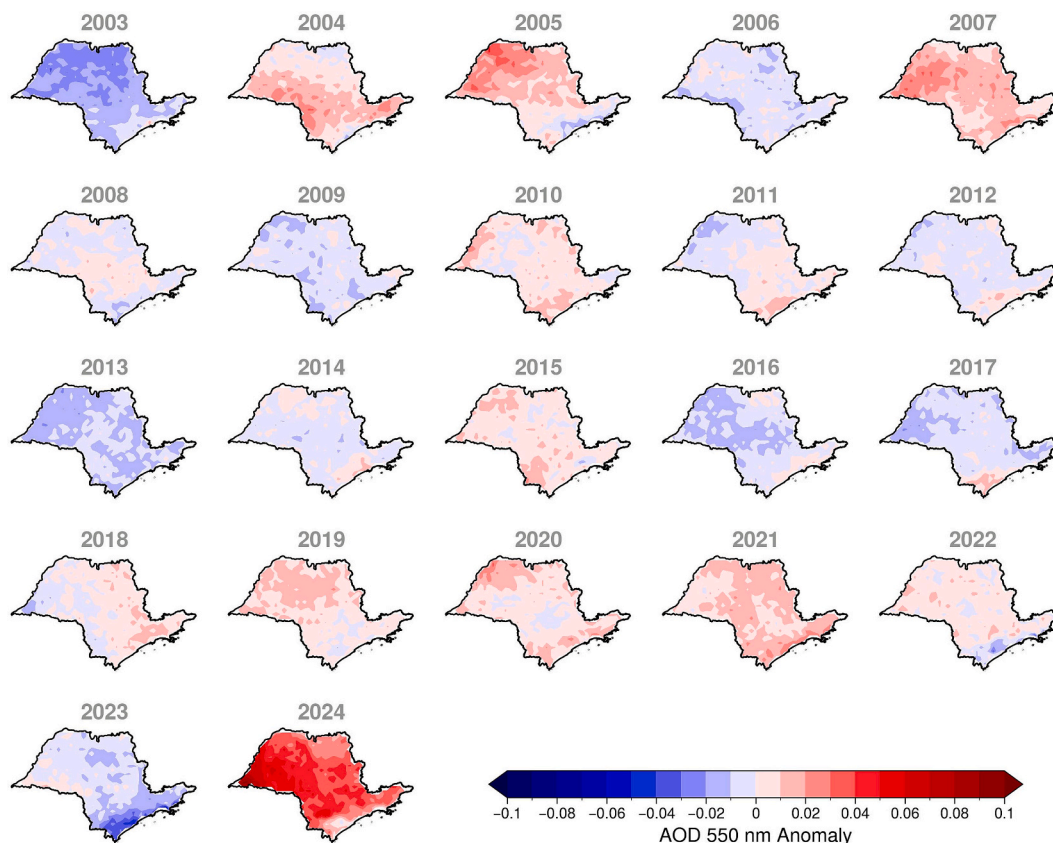


Fig. 9. Annual Aerosol Optical Depth (AOD) 550 nm anomalies in the São Paulo state.

addition to demonstrating a consistent and recurring physicochemical process over the years.

### 3.4. Local hotspots and public health

Spatially, the impacts of the 2024 fire season were not uniform, with the accumulated fire map for August and September (Fig. 10) showing a heavy concentration in the northern and northeastern regions of the state. Among the air quality data, four sites are highlighted in red and were analyzed separately and in detail (Supplementary Figs. S7–S10). Within the hotspots, the cities of Ribeirão Preto and São José do Rio Preto stand out, recording the highest PM<sub>10</sub> concentrations during the period. The graphs (S7 and S8) show that the maximum peak occurred shortly after a compound extreme episode on August 24, exceeding 200 µg/m<sup>3</sup>. The following day, August 25, precipitation significantly reduced pollutant concentrations. Similar results were found by Rudke et al. (2025), who identified maximum PM<sub>25</sub> concentrations on the same date for these cities. The WHO guideline for the 24-h mean of PM<sub>10</sub> is 45 µg/m<sup>3</sup> (WHO – World Health Organization, 2021). On August 24th, however, recorded values reached 200 µg/m<sup>3</sup>, exceeding the recommended daily limit by more than four times and indicating unhealthy air quality conditions. Under pollution levels of this magnitude, avoiding outdoor activities is strongly advised, since PM<sub>10</sub> exposure is directly associated with increased morbidity and mortality related to respiratory and cardiovascular diseases. These effects occur across the entire population but disproportionately impact children, older adults, and individuals with pre-existing health conditions (dos Santos et al., 2025). In contrast, Jaú and Jundiá (S9 and S10) had fewer fire outbreaks and, consequently, slightly lower PM<sub>10</sub> concentrations.

For O<sub>3</sub>, the four cities analyzed recorded higher concentrations during composite extreme events (Figs. S7–S10). Higher temperatures favor the formation of this pollutant, which is evident in the graphs. The city of Jundiá presented the highest value, reaching 189.1 µg/m<sup>3</sup>, almost double the limit recommended by WHO guidelines (100 µg/m<sup>3</sup>, WHO – World Health Organization, 2021), indicating significant degradation of air quality. Exposure to O<sub>3</sub> is associated with an increased risk of cardiovascular and respiratory diseases, as well as higher mortality, especially among vulnerable populations (Kim et al., 2024; Kelley et al., 2025).

The cities of Ribeirão Preto and São José do Rio Preto probably presented the highest concentrations of PM<sub>10</sub> because they are located in areas of intense agricultural activity focused on sugarcane cultivation, where pre-harvest burning is still recurrent (Ogura et al., 2022; Zheng et al., 2022). Although to a lesser extent, the other cities also recorded PM<sub>10</sub> peaks, which may be associated with the transport of smoke plumes from biomass burning areas in neighboring regions (Cristale et al., 2012; Silva et al., 2025; Rudke et al., 2025). This phenomenon highlights the rural-urban interface, in which emissions from agricultural burning directly impact air quality and the health of the urban population (Cristale et al., 2012; Paraíso and Gouveia, 2015; Scaramboni et al., 2024; Rudke et al., 2025). Furthermore, the transport of these plumes contributes to the increased concentration of secondary pollutants, such as O<sub>3</sub>, by intensifying photochemical reactions in the atmosphere (Silva et al., 2025), contributing to the high levels of pollutants observed in all cities.

Beyond the atmospheric impacts, the effects of the wildfires directly impacted the health of the population. This severe environmental degradation in specific regions translated into an alarming public health emergency. A study by Moreira et al. (2025) found that, between January and September 2024, hospitalizations for respiratory and cerebrovascular diseases in São Paulo were significantly higher than the 19-year mean, with children and the elderly being the most affected. Corroborating this finding, Rudke et al. (2025) identified a sharp increase in preventable deaths associated with air pollution during this period. In municipalities heavily impacted by smoke from the fires, such as Ribeirão Preto and São José do Rio Preto, the rates reached approximately 70 deaths per 100,000 inhabitants, values comparable to those observed during critical periods of the COVID-19 pandemic. This comparison, used by the authors to gauge the magnitude of the crisis, shows that the impacts of the fires on public health were severe and large-scale. The growing demand for hospital care overwhelmed the public health system, increasing costs and negatively impacting the local economy. These findings reinforce that the environmental crisis had serious, measurable, and alarming consequences for the well-being of the population living in and around the fire outbreaks.

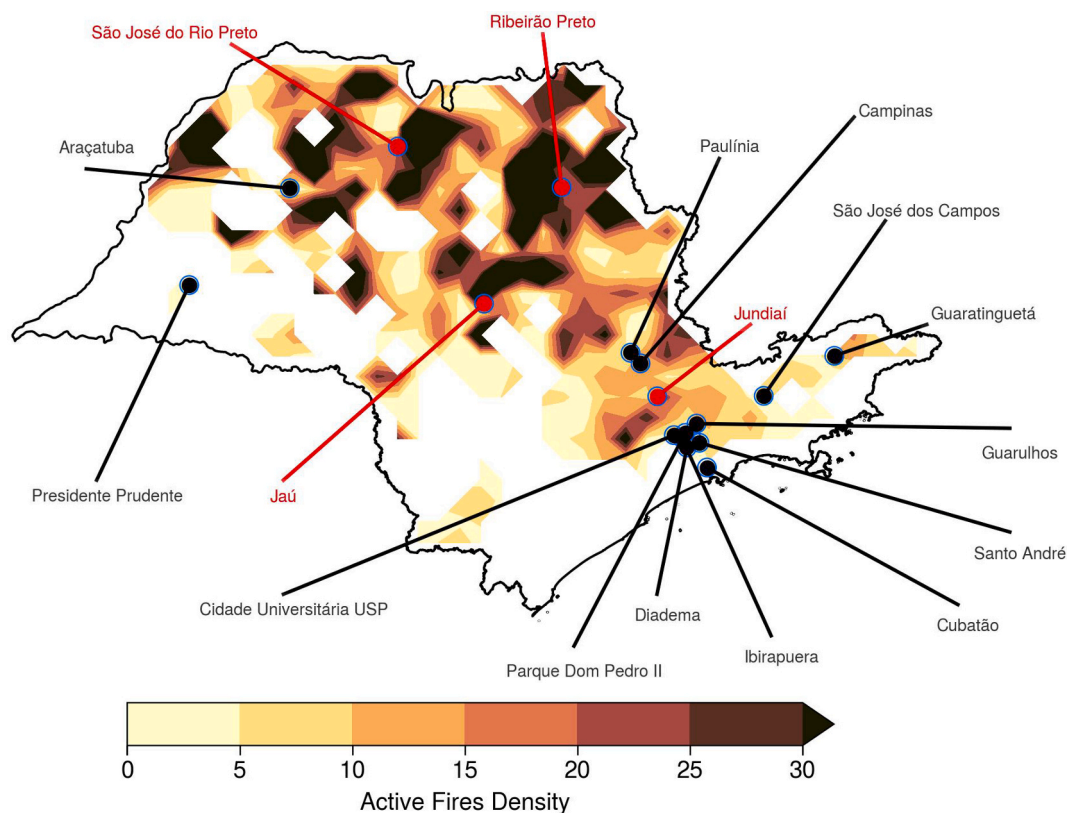
## 4. Conclusion

The 2024 fire season in São Paulo was an exceptional episode that illustrates how compound heatwave–drought conditions can amplify fire activity and air quality degradation, even under a long-term decline in fire outbreaks. São Paulo recorded 8712 active fires in 2024, representing a 118% increase relative to the historical mean and the highest value since 2003. During the most intense compound drought–heatwave periods, PM<sub>10</sub> concentrations exceeded 200 µg/m<sup>3</sup> in some regions, while O<sub>3</sub> concentrations reached approximately 189.1 µg/m<sup>3</sup>. Tmax showed strong positive correlations with PM<sub>10</sub> ( $r = 0.79$ ) and O<sub>3</sub> ( $r = 0.87$ ), reinforcing the important role of compound climate extremes in intensifying fire activity and air quality degradation.

Although the long-term record shows a decreasing trend in active fires since 2003, the 2024 episode demonstrates that severe fire seasons can still occur when exceptional heat and precipitation deficits coincide. Therefore, 2024 should not be interpreted as evidence of a regional increasing trend in fire activity, but rather as a warning of the potential cascading impacts produced by the concurrence of heatwaves, drought, fire outbreaks, and degraded air quality.

This study does not assess future climate scenarios for São Paulo. However, its findings are consistent with broader evidence indicating that compound heat and drought events are expected to become more frequent and/or intense under continued climate change. In densely populated and industrialized regions, such events may substantially increase population exposure and public health risks, particularly among socially vulnerable groups, children, and the elderly.

Risk management strategies should therefore incorporate predictive approaches, including integrated monitoring of compound heatwave–drought events, fire risk, and pollutant dispersion. Future studies should expand the analysis to other drought and fire years



**Fig. 10.** Accumulated number of active fires between august and september 2024 in São Paulo state, with the location of air quality stations. Stations highlighted in red were analyzed in detail. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in São Paulo and include additional fire metrics, such as Fire Radiative Power and burned area, to better characterize fire intensity and its impacts on air quality.

#### CRediT authorship contribution statement

**Paola do Nascimento Silva:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vanessa Silveira Barreto Carvalho:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Aline Araújo de Freitas:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Enrique Vieira Mattos:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.uclim.2026.103045>.

## Data availability

The authors are unable or have chosen not to specify which data has been used.

## References

- Botelho, E.D. do N., dos Santos, A.L.B., de Carvalho, C.R., Pereira, F.G., Dias, T.A.B., 2025. Ocorrências de Incêndios Florestais em Portugal e Brasil, sob a Ótica do Corpo de Bombeiros Militar de Minas Gerais, no Contexto das Mudanças Climáticas. Encontro Latino-americano e Europeu sobre Edificações e Comunidades Sustentáveis (euroELECS), pp. 1–10, 6.
- Bourgoin, C., Beuchle, R., Branco, A., Carreiras, J., Ceccherini, G., Oom, D., Sedano, F., 2025. Extensive fire-driven degradation in 2024 marks worst Amazon forest disturbance in over 2 decades. *Biogeosciences* 22 (19), 5247–5256.
- Cachola, C.S., Andrade, A.C.A.C., Lopes, L.S., Santos, E.M., Peyerl, D., 2022. O papel das políticas públicas na redução das emissões veiculares de gases de efeito estufa no estado de São Paulo. *Desenvolv. Meio Ambiente* 59. <https://doi.org/10.5380/dma.v59i0.76639> e-ISSN 2176-9109.
- Camara, M.R.G.D., Caldarelli, C.E., 2016. Expansão canavieira e o uso da terra no estado de São Paulo. *Estud. Av.* 30, 93–116. <https://doi.org/10.1590/S0103-40142016.30880008>.
- Cantamissa, M.L.D., Satler, L.D., da Silva, M.G.R., da Silva, J.S., 2024. Análise da emissão dos materiais particulados em Manhuaçu-MG e sua correlação com atendimentos por doenças respiratórias. *Caderno Pedag.* 21 (7), e5522.
- Cardoso, T.D.F., Cavalett, O., Chagas, M.F., Moraes, E.R.D., Carvalho, J.L.N., Franco, H.C.J., Bonomi, A., 2013. Technical and economic assessment of trash recovery in the sugarcane bioenergy production system. *Sci. Agric.* 70, 353–360.
- CEMADEN – Centro Nacional de Monitoramento e Alertas de Desastres Naturais, 2024. Boletim de Monitoramento de Secas e Impactos no Brasil – Setembro/2024. CEMADEN, São José dos Campos. [https://www.gov.br/cemaden/pt-br/assuntos/monitoramento/monitoramento-de-seca-para-o-brasil/monitoramento-de-secas-e-impactos-no-brasil-2013-agosto-2024/Boletim\\_secas\\_092024.pdf](https://www.gov.br/cemaden/pt-br/assuntos/monitoramento/monitoramento-de-seca-para-o-brasil/monitoramento-de-secas-e-impactos-no-brasil-2013-agosto-2024/Boletim_secas_092024.pdf) (Accessed 03 Jun 2025).
- CETESB - Companhia Ambiental do Estado de São Paulo, 2024. Qualidade do ar no Estado de São Paulo 2023. <https://cetesb.sp.gov.br/ar/wp-content/uploads/sites/28/2024/08/Relatorio-de-Qualidade-do-Ar-no-Estado-de-Sao-Paulo-2023.pdf>. Accessed 02 May 2025.
- CETESB - Companhia Ambiental do Estado de São Paulo, 2025. Qualidade do ar no Estado de São Paulo 2024. <https://www.cetesb.sp.gov.br/dx/api/dam/v1/collections/186909e9-ab59-4641-abba-c5c465793216/items/49772306-f324-47b6-9c89-a5aa2c8738e6/renditions/201c1df2-d591-4663-a3f7-714bb86f2d50?binary=true>. Accessed 15 Sep 2025.
- Chen, M., Shi, W., Xie, P., Silva, V.B., Kousky, V.E., Wayne Higgins, R., Janowiak, J.E., 2008. Assessing objective techniques for gauge-based analyses of global daily precipitation. *J. Geophys. Res. Atmos.* 113 (D4).
- CNA – Confederação da Agricultura e Pecuária do Brasil, 2024. Impactos dos Incêndios no Setor Agropecuário. CAN, Brasília. Nota Técnica. <https://drive.google.com/file/d/1y2Hx6dCIYzHpAUjNeHcGVRT5JrriGzf/view> (Accessed 01 Jul 2025).
- CNI – Confederação Nacional da Indústria, 2023. Perfil da Indústria – São Paulo. <https://perfildaindustria.portaldaindustria.com.br/estado/sp> (Accessed 01 Jul 2025).
- CNM – Confederação Nacional de Municípios, 2021. Quaimadas Somam Prejuízos De Mais de R\$1,1 Bilhão no Brasil nos Últimos Seis Anos. <https://cnm.org.br/comunicacao/noticias/estudo-cnm-queimadas-somam-prejuizos-de-mais-de-r-1-1-bilhao-no-brasil-nos-ultimos-seis-anos> (Accessed 01 Jul 2025).
- Coelho, T.F., de Souza, A.R., de Lima, J.A.P.L., Pedroza, M.M., 2023. Impactos dos poluentes resultantes das Queimadas na Saúde humana: impacts of pollutants resulting from burns on human health. *Rev. Eng. Tecnol.* 15 (1).
- Cristale, J., Silva, F.S., Zocolo, G.J., Marchi, M.R.R., 2012. Influence of sugarcane burning on indoor/outdoor PAH air pollution in Brazil. *Environ. Pollut.* 169, 210–216.
- Cui, L., Zhu, C., Zou, Z., Yao, C., Zhang, C., Li, Y., 2023. The spatiotemporal characteristics of wildfires across Australia and their connections to extreme climate based on a combined hydrological drought index. *Fire* 6 (2), 42.
- de Moura, F.R., Machado, P.D.W., Ramires, P.F., Tavela, R.A., Carvalho, H., da Silva Júnior, F.M.R., 2024. In the line of fire: analyzing burning impacts on air pollution and air quality in an Amazonian city, Brazil. *Atm. Pollut. Res.* 15 (4), 102033.
- de Sousa, J.A.P., da Silva, R.D.C.F., Duarte, M.L., Lourenço, R.W., 2024. Análise da Ocorrência de Incêndios e Queimadas Mediante as Condições Climáticas e Interferências Antrópicas.
- de Souza, B.F., de Lima, T.L., 2025. Relação entre os incêndios e a legislação ambiental brasileira: reflexão sobre a gestão ambiental e eficácia das políticas públicas implementadas. *NATIVA-Rev. Ciênc. Tecnol. Inovaç.* 8 (1), 203–218 (ISSN: 2764-1295).
- De Souza, A.F., Alves, V.D.S., Vargas, A.N.D.M., Bittencourt, W.S., 2024. O Impacto da queima de biomassa por incêndios florestais nas internações hospitalares e mortalidade: revisão de escopo. *Ensaios e Ciênc. Biol. Agrár. Saúde* 28 (5), 573–580.
- Diniz, F.R., Gonçalves, F.L.T., Sheridan, S., 2020. Heat wave and elderly mortality: historical analysis and future projection for metropolitan region of São Paulo, Brazil. *Atmosphere* 11 (9), 933.
- dos Santos, J.F.L., Kovalsky, B., Ferreira, T.D.S., Batista, A.C., Tetto, A.F., 2022. Adjustment of the grass fuel moisture code for grasslands in southern Brazil. *Fire* 5 (6), 209.
- dos Santos, D.R., Silva, R.S., da Silva Sales, D., Guimarães, H.A., Sales, C.M.R., 2025. Poluição atmosférica e saúde pública: evidências sobre o material particulado nos estados de São Paulo e Rio de Janeiro. *Rev. Polit. Públi. Cidades* 14 (8), e2603.
- Du, H., Yan, M., Liu, X., Zhong, Y., Ban, J., Lu, K., Li, T., 2024. Exposure to concurrent heatwaves and ozone pollution and associations with mortality risk: a nationwide study in China. *Environ. Health Perspect.* 132 (4), 047012.
- Engdaw, M.M., Steiner, A.K., Hegerl, G.C., Ballinger, A.P., 2023. Attribution of observed changes in extreme temperatures to anthropogenic forcing using CMIP6 models. *Weather. Clim. Extremes* 39, 100548.
- Espindola, G.M., 2024. Calor, seca, fogo e fumaça: desastres climáticos e a proteção de vulneráveis no Brasil. *Revista CNJ* 8 (2), 57–76. <https://doi.org/10.54829/revistacnj.v8i2.652>.
- Fernandes, D.S., Heinemann, A.B., da Paz, R.L., Amorim, A.D.O., Cardoso, A.S., 2009. Índices Para a Quantificação da Seca.
- Forster, P.M., Smith, C., Walsh, T., Lamb, W.F., Lamboll, R., Cassou, C., Zhai, P., 2025. Indicators of global climate change 2024: annual update of key indicators of the state of the climate system and human influence. *Earth Syst. Sci. Data* 17 (6), 2641–2680.
- Freitas, A.A., Reboita, M.S., Carvalho, V.S.B., Drumond, A., Ferraz, S.E.T., da Silva, B.C., da Rocha, R.P., 2023. Atmospheric and oceanic patterns associated with extreme drought events over the Paraná hydrographic region, Brazil. *Climate* 11 (1), 12. <https://doi.org/10.3390/cli11010012>.
- Gao, Y., Zhang, L., Zhang, G., Yan, F., Zhang, S., Sheng, L., Gao, H., 2020. The climate impact on atmospheric stagnation and capability of stagnation indices in elucidating the haze events over North China plain and Northeast China. *Chemosphere* 258, 127335.

- Gao, Y., Wu, Y., Guo, X., Kou, W., Zhang, S., Leung, L.R., Wu, L., 2023. More frequent and persistent heatwaves due to increased temperature skewness projected by a high-resolution earth system model. *Geophys. Res. Lett.* 50 (18), e2023GL105840.
- Garcia, L.C., Szabo, J.K., de Oliveira Roque, F., Pereira, A.D.M.M., da Cunha, C.N., Damasceno-Júnior, G.A., Ribeiro, D.B., 2021. Record-breaking wildfires in the world's largest continuous tropical wetland: integrative fire management is urgently needed for both biodiversity and humans. *J. Environ. Manag.* 293, 112870.
- Garrido, B.A., 2024. Mais de 80% dos Focos de Calor em SP Ocorreram Em Áreas Agropecuárias. Instituto de Pesquisa Ambiental da Amazônia (IPAM). <https://ipam.org.br/focos-de-calor-sp/> (Accessed 12 Nov 2025).
- Gavidia-Calderón, M., Schuch, D., Vara-Vela, A., Inoue, R., Freitas, E.D., Albuquerque, T.T.D.A., Bell, M.L., 2024. Air quality modeling in the metropolitan area of São Paulo, Brazil: A review. *Atmos. Environ.* 319, 120301.
- Giglio, L., Schroeder, W., Justice, C.O., 2016. The collection 6 MODIS active fire detection algorithm and fire products. *Remote Sens. Environ.* 178, 31–41. <https://doi.org/10.1016/j.rse.2016.02.054>.
- Google Earth Engine, 2025. MCD19A2.061: Terra & Aqua MAIAC Land Aerosol Optical Depth Daily 1km. [https://developers.google.com/earth-engine/datasets/catalog/MODIS\\_061\\_MCD19A2\\_GRANULES?hl=pt-br#description](https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MCD19A2_GRANULES?hl=pt-br#description) (Accessed 01 Jul 2025).
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google earth engine: planetary-scale geospatial analysis for everyone. *Remote Sens. Environ.* 202, 18–27. <https://doi.org/10.1016/j.rse.2017.06.031>.
- Gouveia, C.M., Bistinas, L., Liberato, M.L., Bastos, A., Koutsias, N., Trigo, R., 2016. The outstanding synergy between drought, heatwaves and fuel on the 2007 southern Greece exceptional fire season. *Agric. For. Meteorol.* 218, 135–145. <https://doi.org/10.1016/j.agrformet.2015.11.023>.
- Hu, A., Xie, X., Gong, K., Hou, Y., Zhao, Z., Hu, J., 2022. Assessing the impacts of climate change on meteorology and air stagnation in China using a dynamical downscaling method. *Front. Environ. Sci.* 10, 894887.
- IBGE - Instituto Brasileiro de Geografia e Estatística, 2024. Cidades e Estados. <https://www.ibge.gov.br/cidades-e-estados/sp.html>. Accessed 02 May 2025.
- IBGE - Instituto Brasileiro de Geografia e Estatística, 2021. Produto Interno Bruto - PIB. <https://www.ibge.gov.br/explica/ PIB.php>. Accessed 12 April 2025.
- IBGE - Instituto Brasileiro de Geografia e Estatística, 2022. População. <https://censo2022.ibge.gov.br/panorama/indicadores.html?localidade=BR&tema=5> (Accessed 12 April 2025).
- INPE - Instituto Nacional de Pesquisas Espaciais, 2024a. InfoQueima: Boletim Mensal de Monitoramento e Risco de Queimadas e Incêndios Florestais. Agosto de 2024a. [https://dataserver-coids.inpe.br/queimadas/queimadas/Infoqueima/2024/2024\\_08\\_infoqueima.pdf](https://dataserver-coids.inpe.br/queimadas/queimadas/Infoqueima/2024/2024_08_infoqueima.pdf) (Accessed 03 Jun 2025).
- INPE - Instituto Nacional de Pesquisas Espaciais, 2024b. InfoQueima: Boletim Mensal De Monitoramento E Risco de Queimadas e Incêndios Florestais [https://dataserver-coids.inpe.br/queimadas/queimadas/Infoqueima/2024/2024\\_09\\_infoqueima.pdf](https://dataserver-coids.inpe.br/queimadas/queimadas/Infoqueima/2024/2024_09_infoqueima.pdf) (Accessed 03 Jun 2025).
- INPE - Instituto Nacional de Pesquisas Espaciais, 2025a. Programa Queimadas - Situação Atual: Estatísticas por Países. <https://terrabrasilis.dpi.inpe.br/queimadas/situacao-atual/estatisticas/estatisticas-paises/> (Accessed 01 Jul 2025b).
- INPE - Instituto Nacional de Pesquisas Espaciais, 2025b. Programa Queimadas: Banco de Dados de Focos de Queimadas. [https://dataserver-coids.inpe.br/queimadas/queimadas/focos/csv/anoal/Brasil\\_sat\\_ref/](https://dataserver-coids.inpe.br/queimadas/queimadas/focos/csv/anoal/Brasil_sat_ref/) (Accessed 01 Ago 2025c).
- INPE - Instituto Nacional de Pesquisas Espaciais, 2025c. Perguntas Frequentes - FAQ. [https://data.inpe.br/queimadas/portal/pages/secao\\_informacoes/faq/index.html](https://data.inpe.br/queimadas/portal/pages/secao_informacoes/faq/index.html) (Accessed 07 Nov 2025d).
- IPCC, 2023. Sections. In: Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team. IPCC, Geneva, Switzerland, pp. 35–115. <https://doi.org/10.59327/IPCC/AR6-9789291691647>.
- Jiménez, J.C., Miranda, V., Trigo, I., Libonati, R., Albuquerque, R., Peres, L.F., Marengo, J.A., 2024. Vegetation warming and greenness decline across Amazonia during the extreme drought of 2023. *Remote Sens.* 16 (14), 2519.
- Kelley, D.L., Burton, C., Di Giuseppe, F., Jones, M.W., Barbosa, M.L., Brambleby, E., Kolden, C.A., 2025. State of wildfires 2024–2025. *Earth Syst. Sci. Data* 17 (10), 5377–5488.
- Kendall, M.G., 1975. Rank Correlation Methods, 4th edition. Charles Griffin, London.
- Kim, M.S., Lim, Y.H., Oh, J., Myung, J., Han, C., Bae, H.J., Lee, D.W., 2024. Long-term ozone exposure, COPD, and asthma mortality: a retrospective cohort study in the Republic of Korea. *Atmosphere* 15 (11), 1340.
- Kumar, N., 2010. What can affect AOD-PM<sub>2.5</sub> association? *Environ. Health Perspect.* 118 (3), A109–A110.
- Kumar, N., Chu, A., Foster, A., 2007. An empirical relationship between PM<sub>2.5</sub> and aerosol optical depth in Delhi metropolitan. *Atmos. Environ.* 41 (21), 4492–4503.
- Libonati, R., Geirinhas, J.L., Silva, P.S., Russo, A., Rodrigues, J.A., Belém, L.B., Trigo, R.M., 2022a. Assessing the role of compound drought and heatwave events on unprecedented 2020 wildfires in the Pantanal. *Environ. Res. Lett.* 17 (1), 015005. <https://doi.org/10.1088/1748-9326/ac462e>.
- Libonati, R., Geirinhas, J.L., Silva, P.S., Monteiro dos Santos, D., Rodrigues, J.A., Russo, A., Trigo, R.M., 2022b. Drought-heatwave nexus in Brazil and related impacts on health and fires: a comprehensive review. *Ann. N. Y. Acad. Sci.* 1517 (1), 44–62. <https://doi.org/10.1111/nyas.14887>.
- Lourenzani, W.L., Caldas, M.M., 2014. Land use change from the sugar cane expansion in the western region of São Paulo state, Brazil/Mudancas no uso da terra decorrentes da expansão da cultura da cana-de-acucar na regioao oeste do estado de Sao Paulo. *Ciência Rural* 44 (11), 1980–1988. <https://doi.org/10.1590/0103-8478cr20140186>.
- Lu, X., Zhang, L., Shen, L., 2019. Meteorology and climate influences on tropospheric ozone: a review of natural sources, chemistry, and transport patterns. *Curr. Pollut. Rep.* 5 (4), 238–260.
- Lyapustin, A., & Wang, Y. (2022). <i>MODIS/Terra+Aqua Land Aerosol Optical Depth Daily L2G Global 1km SIN grid V061</i> [data set]. NASA Land Processes Distributed Active Archive Center. doi:<https://doi.org/10.5067/MODIS/MCD19A2.061> Date Accessed: 2025-11-11.
- Mann, H.B., 1945. Nonparametric tests against trend. *Econometrica* 245–259.
- MAPBIOMAS, 2025. Fogo - MapBiomass. <https://plataforma.brasil.mapbiomas.org/hotsite-fogo> (Accessed 01 Jul 2025).
- Marengo, J.A., Cunha, A.P., Cuartas, L.A., Deusdará Leal, K.R., Broedel, E., Seluchi, M.E., Bender, F., 2021. Extreme drought in the Brazilian Pantanal in 2019–2020: characterization, causes, and impacts. *Front. Water* 3, 639204.
- Marengo, J.A., Cunha, A.P., Luiz, R., Cuartas, A., Pimentel, A., Alvalá, R.C.S., Seluchi, M.E., Broedel, E., Muñoz, V., 2025. Estado do Clima, Extremos de Clima e Desastres no Brasil Em 2024 [Nota técnica]. INCT-MC2 / CEMADEN. [https://inctmc2.cemaden.gov.br/wp-content/uploads/2025/03/Relatorio\\_NT-Clima\\_Extremos\\_Desastres-2024-Brasil-digital.pdf](https://inctmc2.cemaden.gov.br/wp-content/uploads/2025/03/Relatorio_NT-Clima_Extremos_Desastres-2024-Brasil-digital.pdf).
- McKee, T.B., Doesken, N.J., Kleist, J., 1993. The relationship of drought frequency and duration to time scales. In: Proceedings of the 8th Conference on Applied Climatology, Vol. 17, pp. 179–183.
- Melo, K.D.S., Delgado, R.C., Pereira, M.G., Ortega, G.P., 2024. The consequences of climate change in the Brazilian Western Amazon: a new proposal for a fire risk model in Rio Branco, Acre. *Forests* 15 (1), 211.
- Miguel, G.C., Mattos, E.V., Reboita, M.S., Pons, N.A.D., de Oliveira Neto, M.A., 2025. Caracterização das Queimadas no Município de Itajubá, MG. *Rev. Bras. Geogr.* 18 (2), 880–899. <https://doi.org/10.26848/rbgf.v18.2.p880-899>.
- Moreira, G.D.A., Carbone, S., Guerrero-Rascado, J.L., Andrade, I.D.S., Cacheffo, A., Vélez-Pereira, A.M., Diaz, T.P., 2025. Evidence of the consequences of the prolonged fire season on air quality and public health from 2024 São Paulo (Brazil) data. *Sci. Rep.* 15 (1), 28337.
- Moss, W.E., Crausbay, S.D., Rangwala, I., Wason, J.W., Trauernicht, C., Stevens-Rumann, C.S., Booth, R.K., 2024. Drought as an emergent driver of ecological transformation in the twenty-first century. *BioScience* 74 (8), 524–538.
- NOAA. Physical Sciences Laboratory. Gridded Climate Data. <https://psl.noaa.gov/data/gridded/index.html>. Accessed 15 Dec 2024.
- Ogura, A.P., da Silva, A.C., Castro, G.B., Espíndola, E.L.G., da Silva, A.L., 2022. An overview of the sugarcane expansion in the state of São Paulo (Brazil) over the last two decades and its environmental impacts. *Sustainable Prod. Consumption* 32, 66–75. <https://doi.org/10.1016/j.spc.2022.04.010>.
- Olmo, M.E., Weber, T., Teichmann, C., Bettolli, M.L., 2022. Compound events in South America using the CORDEX-CORE Ensemble: current climate conditions and future projections in a global warming scenario. *J. Geophys. Res. Atmos.* 127 (21), e2022JD037708.
- Palmeiro-Silva, Y., Llerena-Cayo, C., Blanco-Villafuerte, L., Rojas-Rueda, D., Arriagada, N.B., Vela-Clavo, Z., Hartinger, S.M., 2025. The 2024 South America ablaze: health impacts and policy imperatives for protecting population health in an era of wildfires. *Lancet Reg. Health Am.* 48.

- Pan, K., Faloona, I.C., 2022. The impacts of wildfires on ozone production and boundary layer dynamics in California's Central Valley. *Atmos. Chem. Phys.* 22 (14), 9681–9702.
- Paraiso, M.L.D.S., Gouveia, N., 2015. Health risks due to pre-harvesting sugarcane burning in São Paulo state, Brazil. *Rev. Bras. Epidemiol.* 18, 691–701.
- Ramos, B.M., de Miranda, R.M., 2023. Avaliação sazonal do aerossol e suas propriedades ópticas na Zona Leste de São Paulo. *Rev. Brasil. Climatol.* 33, 438–464.
- Reboita, M.S., Krusche, N., Ambrizzi, T., da Rocha, R.P., 2012. Entendendo o Tempo e o Clima na América do Sul. *Terra Didat.* 8 (1), 34–50.
- Reddy, P.J., Sharples, J.J., Lewis, S.C., Perkins-Kirkpatrick, S.E., 2021. Modulating influence of drought on the synergy between heatwaves and dead fine fuel moisture content of bushfire fuels in the southeast Australian region. *Weath. Clim. Extremes* 31, 100300. <https://doi.org/10.1016/j.wace.2020.100300>.
- Ridder, N.N., Ukkola, A.M., Pitman, A.J., Perkins-Kirkpatrick, S.E., 2022. Increased occurrence of high impact compound events under climate change. *Npj Clim. Atmos. Sci.* 5 (1), 3. <https://doi.org/10.1038/s41612-021-00224-4>.
- Rosan, T.M., Anderson, L.O., Vedovato, L., 2017. Avaliação da origem de focos de calor em anos de extremos climáticos na Amazônia brasileira. *Rev. Bras. Cartogr.* 69 (4), 731–741.
- Rudke, A.P., Martins, L.D., Martins, J.A., Mantoani, M.C., de Fátima Andrade, M., Kumar, P., de Almeida Albuquerque, T.T., 2025. Where there is smoke, there is fire: long- and mid-range biomass burning role on São Paulo's state air quality. *Environ. Pollut.* 384, 126993.
- Russo, S., Dosio, A., Graversen, R.G., Sillmann, J., Carrao, H., Dunbar, M.B., Vogt, J.V., 2014. Magnitude of extreme heat waves in present climate and their projection in a warming world. *J. Geophys. Res. Atmos.* 119 (22), 12–500.
- Santana, J.C.C., Miranda, A.C., Yamamura, C.L.K., Silva Filho, S.C.D., Tambourgi, E.B., Lee Ho, L., Berssaneti, F.T., 2020. Effects of air pollution on human health and costs: current situation in São Paulo, Brazil. *Sustainability* 12 (12), 4875.
- Santana, A.C.A., Junior, J.C.R., Barros, T.H., Rocha, N., Bender, F.D., Cirilo, J.A., de Paiva, A.L.R., 2024. Estudo de tendência de mudanças climáticas, eventos extremos e análise comparativa entre dois municípios de regiões distintas do semiárido pernambucano. *Geografia Ensino & Pesquisa* 28, e86280. <https://doi.org/10.5902/2236499486280>.
- Santos, B.A., da Silva Oliveira, M.D., do Nascimento Silva, P., dos Santos, J.G.M., Mattos, E.V., Reboita, M.S., 2019. Distribuição espaço-temporal dos focos de calor no Estado de Minas Gerais. *Anu. Inst. Geocienc.* 42 (3), 64–84.
- SÃO PAULO (Estado), 2024a. Secretaria de Meio Ambiente, Infraestrutura e Logística. Governo de SP já aplicou mais de R\$25 Milhões em Multas na Operação SP Sem Fogo. SEMIL, São Paulo. <https://semil.sp.gov.br/2024/09/governo-de-sp-já-aplicou-mais-de-r-25-milhoes-em-multas-na-operacao-sp-sem-fogo/> (Accessed 03 Jun 2025).
- SÃO PAULO (Estado), 2024b. Secretaria de Meio Ambiente, Infraestrutura e Logística. Você conhece a Operação São Paulo Sem Fogo? SEMIL, São Paulo. <https://semil.sp.gov.br/educacaoambiental/2024/09/voce-conhece-a-operacao-sao-paulo-sem-fogo/> (Accessed 03 Jun 2025).
- Sarra, S.R., Müllfarth, R.C.K., 2021. Impactos das queimadas da região Centro-Oeste do Brasil sobre as cidades do estado de São Paulo. *Brazil. J. Dev.* 7 (5), 51237–51257. <https://doi.org/10.34117/bjdv.v7i5.30161>.
- Scaramboni, C., Urban, R.C., de Oliveira, D.P., Dorta, D.J., Campos, M.L.A.M., 2024. Particulate matter from a tropical city in Southeast Brazil: impact of biomass burning on polycyclic aromatic compounds levels, health risks, and in vitro toxicity. *Chemosphere* 350, 141072.
- Sen, P.K., 1968. Estimates of the regression coefficient based on Kendall's tau. *J. Am. Stat. Assoc.* 63 (324), 1379–1389.
- Setzer, A., Morelli, F., Souza, J.C., 2019. O banco de dados de queimadas do INPE. *Biodivers. Brasil.* 9 (1), 239. <https://doi.org/10.37002/biodiversidadebrasil.v9i1.1289>.
- Setzer, A.W., Ferreira, N.J., Morelli, F., 2021. O programa queimadas do Inpe. In: *Queimadas e Incêndios Florestais: Mediante Monitoramento Orbital*. Oficina de Texto, São Paulo, pp. 9–21.
- Shi, Z., Jiang, D., Wang, Y., 2024. Spatiotemporal dependence of compound drought–heatwave and fire activity in China. *Weath. Clim. Extremes* 45, 100695. <https://doi.org/10.1016/j.wace.2024.100695>.
- Silva, M.D., Oliveira, M.C.Q.D., Drumond, A., Rizzo, L.V., 2021a. Air pollutants associated with surface meteorological conditions in São Paulo's ABC region. *Rev. Brasil. Ciênc. Ambient.* 56 (3), 459–469.
- Silva, E.M.D., Carvalho, H.C.M., Silva, L.L.D., Barbosa, W.A., 2021b. Registros de queimadas em vegetação (incêndios) e a climatologia da chuvas no estado do ceará: estudo de caso no período de 2015 a 2019. *Rev. Brasil. Meteorol.* 36, 571–577.
- Silva, G.M.D., Rodrigues, M.F., Pelicer, L.S., Moreira, G.D.A., Cacheffo, A., Lopes, F.J.D.S., Landulfo, E., 2025. Long-range plume transport from Brazilian burnings to urban São Paulo: a remote sensing analysis. *Atmosphere* 16 (9), 1022.
- Souto, C.A., Freitas, A.C.V., Martins, G., 2021. Influência das Condições Meteorológicas Atuais e Futuras na Ocorrência de Queimadas e Incêndios Florestais no Estado de Minas Gerais. *Rev. Brasil. Geogr. Física* 14 (5), 2755–2770.
- Sun, Q., Miao, C., Duan, Q., Ashouri, H., Sorooshian, S., Hsu, K.L., 2018. A review of global precipitation data sets: data sources, estimation, and intercomparisons. *Rev. Geophys.* 56 (1), 79–107.
- Sutanto, S.J., Vitolo, C., Di Napoli, C., D'Andrea, M., Van Lanen, H.A., 2020. Heatwaves, droughts, and fires: exploring compound and cascading dry hazards at the pan-European scale. *Environ. Int.* 134, 105276. <https://doi.org/10.1016/j.envint.2019.105276>.
- Tajudin, M.A.B.A., Madaniyazi, L., Seposo, X., Sahani, M., Tobías, A., Latif, M.T., Ng, C.F.S., 2024. Short-term associations of PM10 attributed to biomass burning with respiratory and cardiovascular hospital admissions in peninsular Malaysia. *Int. J. Epidemiol.* 53 (4), dyae102.
- Tarín-Carrasco, P., Augusto, S., Palacios-Peña, L., Ratola, N., Jiménez-Guerrero, P., 2021. Impact of large wildfires on PM 10 levels and human mortality in Portugal. *Nat. Hazards Earth Syst. Sci.* 21 (9), 2867–2880.
- Veras, M.M., 2024. Mudanças climáticas e incêndios florestais: implicações sobre a saúde. *Ciênc. Cult.* 76 (3), 01–07.
- Vieira, A.C.M., Aragão, F.V., Freire, G.B., Ferreira, M.D., Alexandre, W.N., 2024. As Sanções Penais no Combate às Queimadas no Brasil. In: *Forum Rondoniense de Pesquisa (Vol. 5, No. 10)*.
- WHO – World Health Organization, 2021. WHO Global air Quality Guidelines. Particulate Matter (PM<sub>2.5</sub> and PM<sub>10</sub>), Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide.
- Wikuats, C.F.H., Nogueira, T., Squizzato, R., de Freitas, E.D., Andrade, M.D.F., 2023. Health risk assessment of exposure to air pollutants exceeding the new WHO air quality guidelines (AQGs) in São Paulo, Brazil. *Int. J. Environ. Res. Public Health* 20 (9), 5707.
- Wilks, D.S., 2019. *Statistical Methods in the Atmospheric Sciences*. Academic Press.
- WMO – World Meteorological Organization, 2012. *Standardized Precipitation Index User Guide*, 1090. World Meteorological Organization.
- WMO – World Meteorological Organization, 2023. *Atlas of Mortality and Economic Losses from Weather, Climate and Water-related Hazards (1970–2021)*. <https://wmo.int/publication-series/atlas-of-mortality-and-economic-losses-from-weather-climate-and-water-related-hazards-1970-2021>.
- Yang, Q., Yuan, Q., Yue, L., Li, T., Shen, H., Zhang, L., 2019. The relationships between PM<sub>2.5</sub> and aerosol optical depth (AOD) in mainland China: about and behind the spatio-temporal variations. *Environ. Pollut.* 248, 526–535.
- Zheng, Y., dos Santos Luciano, A.C., Dong, J., Yuan, W., 2022. High-resolution map of sugarcane cultivation in Brazil using a phenology-based method. *Earth Syst. Sci. Data* 14, 2065–2080.
- Zittis, G., Lazoglou, G., Hadjinicolaou, P., Lelieveld, J., 2024. Emerging extreme heat conditions as part of the new climate normal. *Theor. Appl. Climatol.* 155 (1), 143–150.
- Zscheischler, J., Seneviratne, S.I., 2017. Dependence of drivers affects risks associated with compound events. *Sci. Adv.* 3 (6), e1700263. <https://doi.org/10.1126/sciadv.1700263>.
- Zscheischler, J., Martius, O., Westra, S., Bevacqua, E., Raymond, C., Horton, R.M., Hurk, B.V.D., AghaKouchak, A., Jézéquel, A., Mahecha, M.D., Maraun, D., Ramos, A.M., Ridder, N.N., Thiery, W., Vignotto, E., 2020. A typology of compound weather and climate events. *Nat. Rev. Earth Environ.* 1 (7), 333–347. <https://doi.org/10.1038/s43017-020-0060-z>.